

Automated Multimedia Transcription and Translation Platform Using Whisper and MarianMT with Interactive AI Integration

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Abstract: *Efficient transcription and translation of multimedia content, of course, go a long way in facilitating accessibility, education, and even content analysis in today's world. Below is an introduction to a novel architecture for such a system, using Whisper for transcription and MarianMT for translation; the architecture developed here was also done with real-time interactivity using Google Gemini AI and powered with Streamlit, among other things, and it had it process different audio/video files, converting them into accurate transcriptions and translations all stored using AWS S3. The transcribed content can be engaged with through a chatbot using interactive features. Our experiments illustrate huge gains in the accuracy of transcriptions and quality translation with significant improvement even in noisy environments.*

Our experiment shows substantial improvements in transcription accuracy as well as translation quality across a number of languages and noisy channels. Whisper handles challenging accent variations and strong interference with good results, and MarianMT delivers the translation with minimal semantic loss. This system will be a worthy tool for accessibility, communication on a global scale, and content analysis due to its versatility, ease of use, and interactive features. Some future work will be focused on improving this support in terms of language by smoothing out the capabilities of real-time interaction.

Keywords: Transcription system, Translation platform, Whisper model, MarianMT translation, Streamlit web app, AWS S3 storage, Google Gemini AI, Real-time interaction, Speech-to-text, Multilingual translation.

I. INTRODUCTION

Real need for accurate transcription and translation of multimedia is rapidly gaining prominence nowadays, especially concerning the areas of accessibility, learning, and content analysis. Paid services from the traditional sources are primarily used, and such access might become nearly impossible for most users seeking immediate solutions. Furthermore, access to these services may become pretty challenging to achieve while working in environments with poor or unreliable internet connectivity. Some of the problems related to access and transcribability of audio files and how it ranks them, because the access to free-of-payment services or stable internet is also a crucial issue in translation; our project solves these problems through the embedding of Whisper for transcription and MarianMT for multilingual translation into a Streamlit- based platform which also provides complimentary free of charge and efficient transcription and translation services in different settings, enabling the effective dealing with audio files.

Whisper is one of the OpenAI models for transcriptions, and coincidentally it is also the best transcription tool found in the noisiest environments with people of the most diverse accents. Using MarianMT with that model, it translates a neural machine translation model over 100 languages, making sure that what is transcribed is also said correctly in its context. Working with the safe cloud-based file storage and retrieval solution based on AWS S3 makes things go more smoothly. Any kind of multimedia material can be uploaded using the upload feature for immediate processing. The technology is more user-friendly and adaptable thanks to integration with Google Gemini AI, which enables users to engage with transcriptions and get contextual responses. As a result, the app is very valuable for education, worldwide communication, and accessibility. Gemini models were trained jointly across image, audio, video,

and text data for the purpose of building a model with both strong generalist capabilities across modalities alongside cutting-edge understanding and reasoning performance in each respective domain.[1] Extensive testing has demonstrated considerable increases in transcribing accuracy in difficult conditions, but most importantly, it offers dependable translations across multiple languages. It has a strong and expandable foundation that not only addresses the shortcomings of the conventional system but also paves the way for future advancements in multimodal content management.

II. PROBLEM STATEMENT

Actually, there are a lot of challenges that make listening to audio recordings after they have been transcribed and translated difficult. Most services are charged for or require a dependable internet connection, which makes it more difficult for customers to access them. The inaccessibility of transcription tools may cause a delay non interpreting crucial content included in audio recordings that carry urgent and time-sensitive information. However, remotely stored data that is maintained by a service provider raises concerns related to the security and privacy of the data.[2] Our initiative provides free, high-quality transcription and translation services, allowing clients to process their data quickly and fully control audio files without being limited by paid services or even needing an internet connection.

III. LITERATURE REVIEW

Over the past few decades, the field of audio transcription and translation has undergone tremendous change due to developments in natural language processing and machine

learning. Traditional transcribing techniques, which frequently involved manual labor, have been progressively replaced by automated systems that use algorithms to boost efficiency. Significant gains in transcription accuracy have been shown by recent developments, especially in deep learning, which allow computers to process a variety of audio inputs, including background noise and accents. The potential of neural networks in this field is demonstrated by studies that demonstrate models such as OpenAI's Whisper can transcribe spoken language with remarkable accuracy.

With neural machine translation (NMT) frameworks like MarianMT providing improved quality through end-to-end learning processes, translation technologies have advanced similarly in parallel. Although these methods have demonstrated potential for high-resource languages, there are still a lot of gaps for low-resource languages, which are frequently under-represented in the research today. This disparity highlights the urgent need for studies aimed at creating efficient transcription and translation tools for a range of linguistic situations. Machine translation (MT) plays a decisive role in cross-lingual information access and will continue to grow as a crucial component in our natural language processing (NLP) toolbox, enabling inclusiveness and equity among people with different cultural and linguistic backgrounds.[3]

The goal of a speech recognition system should be to work reliably 'out of the box' in a broad range of environments without requiring supervised fine-tuning of a decoder for every deployment distribution.[4] The integration of transcription and translation processes is still a significant difficulty despite the developments. Particularly for users that need both services at the same time, many current frameworks handle both jobs as separate, which might result in inconsistencies and inefficiencies. Furthermore, the literature has not sufficiently addressed how transcription errors affect the quality of later translations. Improving the overall user experience and quality of the results requires an understanding of how these processes interact. Furthermore, there is an increasing need for real-time translation and transcription systems that can work in dynamic settings like multilingual meetings or live events. The speed and precision of current solutions are frequently lacking, especially in noisy environments. Furthermore, there are currently few user-centric studies examining how people utilise and assess these tools. Resolving this user experience issues could greatly improve transcription and translation systems' acceptability and usability.

IV. TRANSFORMERS

Transformers represents a significant advancement in machine learning architecture, particularly for natural language processing (NLP) applications like transcription, translation, and text synthesis. By creating what amounts to a novel self-attention mechanism, they have allowed the model to conditionally focus weights on any word or token in the input sequence. The entire sequence is analyzed in parallel rather than sequentially, in contrast to typical models like RNNs, recurrent neural networks, and transformers. This saves little to no computation time and may reveal long-range dependencies in the data.

4.1 Self-Attention Mechanism

When formulating predictions, transformers internally use a method known as self-attention, which enables the model to focus more on other portions of the input sequence. Assume that you are translating sentences. To generate each word in the output sentence, the transformer will determine which terms in the input sentence are most relevant. By calculating different "attention" scores for every word in the input sequence, this is accomplished.

4.2 Encoder-Decoder Architecture

Transformers typically have an encoder decoder architecture:

- Encoder: The encoder processes the input data (e.g., audio or text) by converting it into a continuous representation (embedding). Each layer in the encoder applies self-attention to refine this representation.
- Decoder: The decoder takes this representation and generates an output (e.g., a transcription or translation). Like the encoder, the decoder applies attention mechanisms to both the input and the generated output at each step, ensuring that the context is maintained.

4.3 Multi-Head Attention

Another special ingredient of transformers is multi-head attention, which enables the model to attend to multiple aspects of the input all at once. Multi-head attention is opposed to the classic attention mechanism, where the model only focuses resources on one part of the sequence at a time. It enables the capture of several types of relationships between words or tokens in parallel, which is essential when modeling natural language. Such richness often relies on how the meaning of one word depends on many factors.

4.4 Positional Encoding

Contrasting with traditional sequence models, like RNNs, transformers have no predisposition toward the ordering of tokens in a sequence. Due to this, transformers rely on positional encodings to inject information about a token's position in a sequence. This will help the model know which words come first or last and which are in the middle to maintain the syntactic structure of the language.

4.5 Parallel Processing

One of the strong benefits of working with transformers is parallel processing. While RNNs or LSTMs can all take one token at a time, transformers process an entire sequence in one go. All this makes them quite efficient, especially on big datasets, and able to handle longer sequences without the loss of context.

V. TRANSCRIPTION AND TRANSLATION MODELS

5.1 Whisper

Whisper is an open, advanced ASR model developed by OpenAI that transcribes spoken language into text. It first uses a transformer-based architecture that has spectacular results on the unconditioned sequence-to-sequence task. This is an enormous difference from all other ASR models, which cannot generalize from one class of speech input to another and are thus not very versatile concerning a wide spectrum of

languages, accents, and qualities of audio.

5.1.1 Training and Preprocessing

Whisper is trained on an exceptionally large dataset of multilingual speech data in diverse settings, noise levels, and accents. Such training makes Whisper strong in recognizing

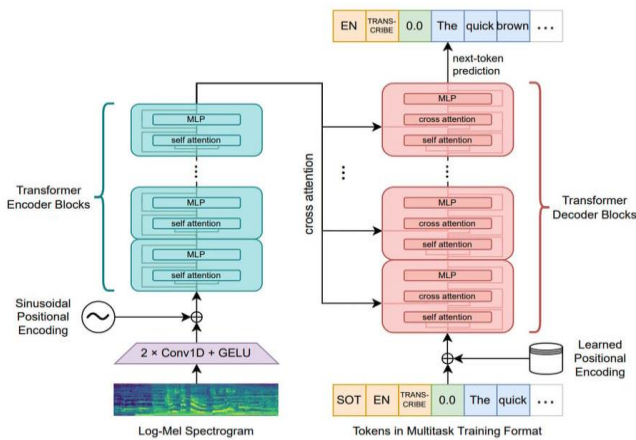


Fig 5.1[4]

speech from linguistically and culturally diverse backgrounds. Furthermore, Whisper can carry out multi-language transcription, enabling it to transcribe languages other than English. It includes spectrogram-based input, whereby audio is converted to a spectrogram that captures the frequency and time information of the speech. This representation then goes to the transformer architecture, which applies self-attention mechanisms to finding the relations of its constituent parts in the audio.

5.1.2 Self-Attention and Sequence-to-Sequence Learning

Whisper has been designed with the transformer model's most salient feature: the self-attention mechanism, which allows it to pay attention to diverse parts of the audio signal. The self-attention mechanism allows the model itself to decide which part of the input is relevant for producing the output text, even if the relevant portions are separated by pauses, noise, or multiple speakers. Thus, Whisper can achieve good transcriptions even for long and complex audio inputs. This is because the sequence-to-sequence learning in Whisper allows it to handle long audio sequences efficiently. Unlike the traditional ASR models that usually do the processing, the analogy of audio being processed in a frame-by-frame manner, Whisper takes an entire audio sequence and outputs the transcription of the said audio.

5.1.3 Capabilities and Robustness

Among many other things, Whisper shines because it takes on real-world noise and various accents without significant degradation in the quality of transcription. The robustness is due to a vast variation in speech data used during training, making such a model fit many applications. Whisper allows multitarget transcription: one single model can transcribe speech in several target languages. Thus, it has proved very versatile for users from all over the world. What is more, Whisper supports fine-tuning for adapting to specific languages or usage scenarios. Using transfer learning techniques, one can further train Whisper on domain-specific datasets to achieve better performance in specialized applications.

5.2 Translation Models

Machine translation (MT) has progressed significantly in recent years, with various models pushing the boundaries in multilingual contexts. Here, we provide an overview of four prominent MT models—MarianMT, M2M100, mBART, and NLLB Distilled—highlighting their architectures, data sources, language coverage, and performance metrics.

5.2.1 MarianMT

The Helsinki-NLP team created MarianMT, a neural machine translation model based on the Marian framework. By training on the OPUS corpus, a large multilingual dataset with more than 1,000 parallel corpora, it leverages the transformer architecture to facilitate translation across many language pairs. MarianMT is extremely extendable, enabling new language pairs or domain-specific fine-tuning, and was designed to strike a compromise between translation efficiency and quality. For high-resource languages like English, French, and Spanish, where extensive parallel data allows for accurate translation, the model's accuracy is robust. With low-resource language pairs, MarianMT's performance may be less reliable because there may be less training data available, which could result in fewer accurate translations. It is widely used for research applications and as a foundation for bespoke programming because of its open-source nature and robust support for high-resource languages. Marian has minimal dependencies (only Boost and CUDA or a BLAS library) and enables barrier-free optimization at all levels: meta-algorithms such as MPI-based multi-node training, efficient batched beam search, compact implementations of new models, custom operators, and custom GPU kernels[5]

5.2.2 M2M 100

Meta's M2M100 model is one of the pioneering multilingual translation models to avoid using English as an intermediary language, offering direct translation between 100 languages. This model leverages the transformer architecture and was trained on a large-scale dataset of 7.5 billion sentence pairs across multiple languages. By training on direct pairs, M2M100 can capture the nuances of less common language combinations, which often require bridging through English in traditional MT setups. M2M100 excels in translating widely spoken languages and performs well with linguistically similar languages, as it can learn shared patterns and structures. However, due to the complexities of fully direct multilingual translation, its performance can vary, particularly with highly divergent languages. Additionally, M2M100's large model size, while beneficial for accuracy, can require considerable computational resources, presenting a trade-off for institutions without access to high-performance computing.

5.2.3 mBART

The mBART model, a multilingual variant of Facebook's BART model, is trained using denoising autoencoding on monolingual data, positioning it for a range of multilingual tasks. Unlike MarianMT and M2M100, mBART is not initially trained on parallel translation data but on large volumes of monolingual text in numerous languages. This training approach helps mBART develop generalizable language representations, which can then be fine-tuned for specific language pairs. The model's denoising objective makes it particularly effective for low-noise, high-quality text generation in target languages, though it sometimes underperforms on direct translations relative to models trained with parallel corpora. mBART's ability to generate natural-sounding text across languages makes it an attractive option for

tasks beyond translation, such as summarization and data augmentation, making it a flexible tool for multilingual natural language processing applications.

5.2.4 NLLB Distilled

Meta's NLLB (No Language Left Behind) project focuses on providing quality translation for low-resource languages, an area often neglected in mainstream MT development. The NLLB Distilled model is a lightweight version of the original NLLB, designed to provide reliable translations with reduced computational requirements. This model is trained on a vast, diverse multilingual corpus, with targeted attention on languages that frequently lack digital resources. NLLB Distilled shows impressive performance on low-resource language pairs, especially those with limited existing translation data. While it may encounter challenges with extremely low-resource languages where minimal text data is available, NLLB Distilled marks a significant step toward inclusivity in language technology, helping bridge the gap in accessibility for underrepresented languages. Its smaller model size, optimized for efficiency, makes it well-suited for applications where computational resources are limited.

VI. GOOGLE GEMINI API

The Google Gemini API makes use of this language's capacity to provide natural language generation, comprehension, and processing for applications. Advanced machine learning models, such as deep learning, are utilised to perform natural language processing (NLP) tasks like question answering, language comprehension, and conversation production. Gemini 1.5 models are built to handle extremely long contexts; they have the ability to recall and reason over fine-grained information from up to at least 10M tokens.[6] Because of its size, one can interface in real time thanks to the pre-trained models.

6.1 Natural Language Processing Capabilities

The Google Gemini API is powered by a strong algorithm for natural language processing. This involves a model trained on large-size data sets, improving its ability to understand the grammar of a human language. Some of the capabilities include parsing syntax, semantic analysis, and understanding contexts, which help the API understand exactly what the intent of a user could be.

6.2 Multimodal Interaction and Context Awareness

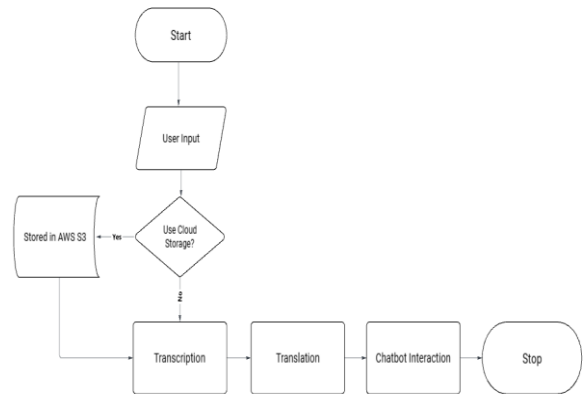
The Gemini API is designed to take in multimodal inputs, meaning it processes not just text but can integrate other forms of data such as audio or images, depending on the application. It also possesses context awareness, with the API maintaining a conversational state that will enable it to keep a memory of and respond to interactions from previous sessions for coherence and continuity in conversations.

VII. SYSTEM ARCHITECTURE

Through the integration of advanced technologies across several layers, our system architecture aims to optimize efficiency and user engagement. This architecture is designed to manage intricate workflows and guarantee data security and integrity all along the way.

Streamlit

The user-facing interface of our application is built on the Streamlit framework, known for its simplicity and effectiveness in creating interactive web applications. Here, users can easily upload audio and video files, which are immediately prepared for processing. The frontend is designed to be intuitive, allowing users to navigate seamlessly through various functionalities, including file upload and real-time viewing of transcription and translation results.



APIs and Models

At the heart of our system lies a robust backend architecture equipped with several advanced machine learning models, each serving a distinct function to enhance the system's overall performance. The Whisper Model takes the lead in transcription, adeptly converting spoken language from uploaded audio files into accurate text. This paves the way for the MarianMT model, which picks up the transcribed text and performs translations into multiple languages. This functionality is crucial in ensuring that our system can cater to a diverse, global user base. Complementing these two is the Google Gemini API, which drives the chatbot interactions. This API interacts dynamically with users by leveraging the transcribed and translated text to provide personalized responses and support, thereby enriching the user experience with engaging and contextually aware interactions.

Cloud Storage Integration

A key feature of our system is the decision node that determines whether to store data in the cloud. Users can choose whether their files should be stored in AWS S3 during the upload process. This flexibility allows for better data management and security, catering to individual user preferences and requirements.

AWS S3

For users opting for cloud storage, AWS S3 is utilized to securely store all uploaded files and processed outputs. AWS S3 is renowned for its scalability and reliability, making it an ideal choice for managing large volumes of data efficiently. This integration ensures that files are not only safely stored but also readily accessible for further processing or retrieval.

Scalability and Maintainability

The entire system is designed to be scalable and maintainable, with modular components that can be independently updated or replaced as technology advances. This modularity allows for rapid deployment of new features and easy adaptation to changing user needs or technological advancements.

VIII.METHODOLOGY

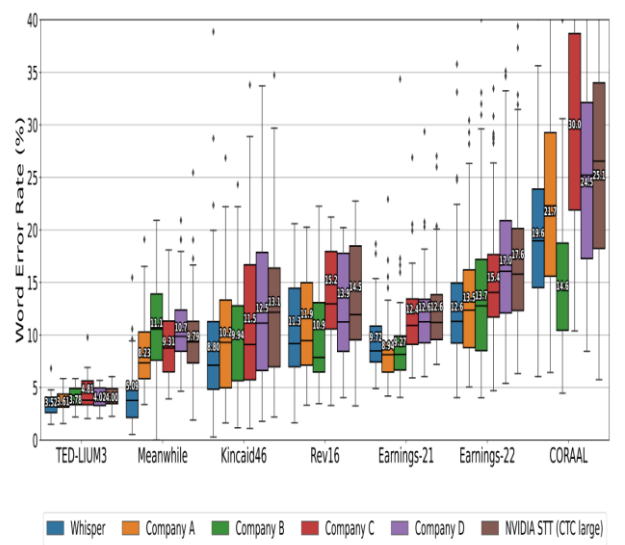
The platform’s core functionality is driven by three major Models:

- Whisper (Transcription): After preprocessing, the audio files undergo transcription within the Whisper model. They generate multi-lingual, high-accuracy results. This stage is very efficient since Whisper has excellent ability in noise handling and letting out a variety of accents.
- MarianMT (Translation): Once the transcription is complete, the text is passed to the MarianMT model if translation is required. MarianMT supports over 100 language pairs, ensuring contextually accurate translations.
- Google Gemini API (Real-Time Interaction): Through this chatbot interface, answers are driven by the Google Gemini API, which responds in real time and takes contextual understanding into account, making the system even more dynamic. After the chatbot has been translated and transcribed, the user can utilize it to ask questions regarding what has been stated.

IX.RESULTS

9.1 Transcription Accuracy

To evaluate the transcription accuracy of the Whisper model, a comparative analysis with Wav2Vec 2.0 (without a language model) is conducted across multiple benchmark datasets, including LibriSpeech, Common Voice, and FLEURS. This analysis will provide insights into the relative performance of these models in terms of their transcription capabilities.



Dataset	wav2vec 2.0 Large (no LM)	Whisper Large V2	RER (%)
LibriSpeech Clean	2.7	2.7	0.0
Artie	24.5	6.2	74.7
Common Voice	29.9	9.0	69.9
Fleurs En	14.6	4.4	69.9
Tedlium	10.5	4.0	61.9
CHiME6	65.8	25.5	61.2
VoxPopuli En	17.9	7.3	59.2
CORAAL	35.6	16.2	54.5
AMI IHM	37.0	16.9	54.3
Switchboard	28.3	13.8	51.2
CallHome	34.8	17.6	49.4
WSJ	7.7	3.9	49.4
AMI SDM1	67.6	36.4	46.2
LibriSpeech Other	6.2	5.2	16.1
Average	29.3	12.8	55.2

Table 9 - Both models show minimal WER on the LibriSpeech Clean dataset, achieving a rate of 2.7%.This indicates that both Whisper and wav2vec 2.0 are highly accurate in ideal, controlled conditions where audio is clear and noise is minimal. In clean environments, the transcription quality of both models is comparable, confirming that each can handle high-quality audio with little to no degradation.

Whisper demonstrates a marked improvement over wav2vec 2.0 in more complex audio environments. For instance, in the Artie dataset, which consists of more challenging speech samples, Whisper achieves a WER of 6.2% compared to wav2vec 2.0's 24.5%. This yields a Relative Error Reduction of 74.7%, highlighting Whisper's ability to accurately transcribe audio with greater noise or variability. Similar patterns are observed in datasets such as Common Voice (RER of 69.9%), Fleurs En (69.9%), and CHiME6 (61.2%), which also feature diverse accents, speaking styles, and background noises. We demonstrate that Distil-Whisper maintains the robustness of Whisper to different audio domains and noisy acoustic conditions.[7]

This improved performance is consistent across most datasets, demonstrating that Whisper is more resilient to factors like background noise, accent variation, and less Fig 9.1[4] structured conversational patterns. Such versatility makes Whisper a more reliable model for real-world scenarios where audio conditions are often suboptimal.

In summary, while both models perform excellently on clean speech, Whisper consistently outperforms wav2vec 2.0 in diverse audio environments. Its lower WER and higher RER across noisy, accented, and conversational datasets underscore Whisper's strength as a general-purpose ASR model, capable of handling a wide range of audio qualities with high accuracy.

The transcription quality of the Whisper model can be assessed through an analysis of its long-form transcription capabilities, as many contemporary models exhibit challenges in this area. A comparative analysis with state-of-the-art models will facilitate a comprehensive evaluation of Whisper’s transcription performance, particularly in long-form contexts.

The long-from transcription is evaluated on 7 datasets consisting of varying recording lengths with varying conditions. Some of

the datasets are long-form adaptations of TED – LIUM3, Rev 16 and Kincaid46.

The performance is compared with 4 commercial ASR services and the NVIDIA STT Conformer-CTC Large model.

Figure 9.1 - Whisper demonstrates strong performance across various long-form datasets. The model consistently outperforms commercial ASR services and the NVIDIA STT Conformer-CTC Large model in terms of Word Error Rate (WER).

Whisper excels in datasets with varying recording lengths and conditions. This highlights its adaptability to diverse audio inputs, making it suitable for various real-world applications. Whisper's performance is particularly impressive in the challenging long-form adaptations of TED-LIUM3, Rev16, and Kincaid46. This indicates its robustness in handling longer audio segments, which is a critical factor in many practical scenarios.

9.2 Translation Quality

In evaluating translation quality across different models, we relied on BLEU and BERTScore metrics, which assess the accuracy and contextual alignment of translations with the source material. The results reveal a notable range in performance, with specific models excelling in different language pairs, while others displayed room for improvement, especially with low-resource languages.

The NLLB-200 Distilled model achieved the highest BLEU score (30.2573) and a strong BERTScore F1 (0.9156), marking it as the top performer overall. This model demonstrated robust translation quality, particularly excelling in contexts requiring nuanced and contextually aligned translations. It effectively supported diverse languages with strong contextual awareness, which reflects its optimization for low-resource languages while maintaining high accuracy in well-resourced ones.

Helsinki-NLP's Opus-MT models performed variably across languages. Opus-MT-en-fr scored a BLEU of 18.5173 and a BERTScore F1 of 0.8779, showcasing its suitability for high-resource European languages, such as French. However, its performance dipped when handling other languages, such as Opus-MT-en-de (BLEU 4.2156, BERTScore 0.9440) and Opus-MT-en-ar (BLEU 3.6058, BERTScore 0.8912). This variance highlights that while the Opus-MT models are well-suited for specific languages, their accuracy may not consistently translate across all language pairs.

Facebook's mBART and m2m100_418M models showed lower BLEU and BERTScore results, with mBART achieving the lowest BLEU (0.0000) and BERTScore F1 (0.6785). The mBART model's performance suggests challenges in maintaining translation fidelity in certain language pairs, particularly in low-resource or non-European languages. In contrast, the m2m100_418M model achieved a higher BERTScore F1 (0.8895) and a BLEU score of 8.1234, indicating a moderately reliable performance with direct multilingual capabilities, particularly beneficial in diverse language applications. The Helsinki-NLP Opus-MT models for Hindi and Chinese also scored low in BLEU (0.0000) with BERTScore F1 scores of 0.7770 and 0.8778, respectively. This

underscores a common difficulty in translating to and from these languages, possibly due to distinct linguistic structures that challenge traditional translation models.

X.CONCLUSION

10.1 Summary of Findings

This research focused on evaluating the performance of various speech recognition and translation models to identify the most effective solutions for transcription and multilingual translation tasks. Our analysis revealed that Whisper Large V2 significantly outperformed wav2vec 2.0 in most datasets, with an average relative error rate (RER) reduction of 55.2%, achieving an average RER of 12.8% compared to wav2vec's 29.3%. Notably, Whisper performed particularly well on challenging datasets such as Artie (RER of 6.2%) and Common Voice (RER of 9.0%), suggesting its robustness in handling varied audio conditions and accents.

In terms of language translation, the NLLB-200 Distilled model demonstrated the highest quality translations across multiple languages, with a BLEU score of 30.2573 and BERTScore F1 of 0.9156, making it the most consistent performer. Other models like MarianMT and Opus-MT provided competitive results in high-resource languages but faced limitations with low-resource language pairs. Meanwhile, models such as mBART showed significant room for improvement, especially in maintaining contextual accuracy across diverse languages.

10.2 Future Work

In future work, our system could benefit from domain-specific fine-tuning and adaptable model-switching to optimize its performance across varied contexts. By training transcription and translation models on specialized datasets for fields like healthcare, legal, or technical domains, we could improve accuracy with complex, field-specific language. Additionally, integrating a dynamic model-switching feature would allow the system to automatically select the best transcription and translation model based on real-time audio characteristics, such as background noise or accent, ensuring consistent accuracy in diverse environments.

Furthermore, exploring cross-modal capabilities and privacy-preserving methods could broaden the system's applicability and safeguard user data. For instance, integrating visual inputs from images or videos could provide additional context for more accurate transcriptions, especially in multimedia applications. Privacy-preserving approaches, like on-device processing or federated learning, could address confidentiality concerns in sensitive areas, such as medical or legal transcription. These enhancements would make the system more versatile, secure, and robust, expanding its potential for use in specialized and privacy-focused applications.

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